ME 350R / ME 380R · ROBOT MECHANISM DESIGN · FALL 2026

Concept Proposal: tell us what your robot will do

Due date: upload to Canvas by Wednesday 30 September. One submission per team. 2–3 pages.

This is the first gate of the main project. It has two halves. First, you describe the robot you want to build — what it is, why you chose it, what it will do on demo day. Then you turn that description into a **specification**: the numbers your robot must meet. The description is what gets you excited; the specification is what you will design to, predict, and be measured against.
A proposal that describes a device but does not say what it must achieve does not pass this gate, and comes back to you.

1. The brief

Your mechanism must engage the physical world through contact — grasping, walking, or worn on a body. What it does with that contact is entirely up to you.

Two rules apply to every project.

Two actuators, maximum. Motors, servos, solenoids — anything that puts energy in counts. A robot that does something impressive with two motors is a mechanism-focused design, while the same thing with sixteen servos is a software demonstration with hardware attached. There is no limit on the number of sensors.

Another way to say it is that **the intellectual content must be in the mechanism**, not in the code. If the interesting part of your project is the program, it is the wrong project for this course.

Choose a track. Name one of the three: *hands and end effectors* · *wearables and data capture* · *legs and locomotion*. Projects that sit between tracks are fine; pick the closest, because it decides where you look for precedent in Section 4.

Creativity, whimsy, and humor on top of engineering excellence are welcome. Some of the best work in this course's archive is also its most playful.

2. The pitch

Write a paragraph for each of the following questions.

What is it? Describe your robot the way you would describe it to a friend who is not an engineer. What does it do, what does it touch, and what happens when it does.

Why this one? What made your team pick it? The joke, the itch, the person it is for, the thing you have always wanted to see move.

Demo day. It is 18 November. Your robot is on the table with the whole class around it. Tell us how you would describe it on that day in thirty seconds. What does it do, in what order, and what do people say when it does?

> Everything you just wrote is a description of the robot. Now read your demo-day paragraph again and pull the numbers out of it. How far does it move? How much does it push? How fast? How many separate things have to move on their own? Those numbers are the specification, and the rest of the semester is built on them: you will synthesize a linkage to hit them in October, predict them in November, and measure them on demo day.

3. The specification

Fill in every row. Each row wants a number or a range, with units. Where you do not know yet, give your best estimate; even a rough estimate is a specification.

Row	*What we are asking*	*Your specification*
What it must move	The thing your mechanism acts on — an object, a foot, a wrist — and what *kind* of motion it needs: a path in space, a set of positions it must reach, a range of angle, a straight line. Sketch the target motion, not the mechanism.	
How far	The size of that motion: travel in mm, angle in degrees, stride, opening width.	
How hard	The force or torque, in N or N·mm: how much, where it is applied, in which direction, and through which part of the motion it is needed.	
How fast	Cycle time, speed, or how often it repeats. Once is a number too.	
How many things at once	The number of independently controlled motions — your degrees of freedom — and therefore your actuator count. Two is the ceiling.	
Mechanics or control	For each requirement above: is it met by the mechanism itself — geometry, springs, compliance, a stop — or by the actuator and its program? One line per row.	

The last row is critical for this course. For instance, a parallelogram mechanism keeps a platform level with no sensor and no code; a compliant finger wraps an unknown object with no model. State the same for your robot.

4. Precedent

The course archive on the wiki holds every project since 2012: https://cloud.wikis.utexas.edu/wiki/spaces/RMD/overview. Find **three** projects there, at least one in your track. For each, four lines:

- what the mechanism was;
- what its analysis claimed it would do;
- what worked on demo day, as far as you can tell from the page and the video;
- what you would change.

Then one sentence: how your proposal differs from all three.

5. What you already know is hard

One short paragraph. The part of this project you are least sure you can build in seven weeks, and what you will try first to find out. Naming it now is helpful because we can focus on that component of the project.

6. A first guess (optional)

If you already have a mechanism in mind, you may include **one** sketch of it, labeled as a first guess, with no dimensions. This part is not graded but it might be a useful way to get started. The mechanism is designed in the synthesis studio in October — from your specification, against the screens you will learn between now and then. Our suggestion is to start with the specification and have an open mind about how the mechanism gets synthesized.

7. How this is graded

Out of 20 points, within the Specify component of the project grade.

The pitch — a demo-day scene specific enough that numbers can be read out of it	4
The specification — a number or a stated estimate in every row, with units	8
Mechanics or control — the split stated for each requirement, and defended in a line	3
Precedent — three projects, at least one in your track, and how yours differs	3
The hard part — named, with a first test	2

> **The gate.** A proposal whose specification has no numbers, or that does not state the mechanics-or-control split, is returned. You have 48 hours to resubmit.

8. What happens next

Once your proposal passes you may begin design and building work. The specification you submit here is the one you will synthesize against in October and register predictions against in November; you may revise it, but every revision is written down. The full design proposal is due at the design freeze on Wednesday 21 October. Demo day is Wednesday 18 November.